

Certified Penetration-Safety Envelopes for Directed-Energy Area Defense against Adversarial Drone Swarms

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github.com/overthelex/hpm-saturation-analysis

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Abstract

High-power-microwave (HPM) weapons are being fielded as the primary counter to drone swarms, on the premise that one “one-to-many” pulse defeats many targets at once. Whether a given emplacement actually *holds* against an adversarial swarm—and against which swarm configurations it fails—is today decided by ad-hoc live-fire demonstrations, not by any principled guarantee. We import and extend the machinery of *certified robustness* from machine learning to physical area defense. Given a defense configuration d , we compute a *certified penetration-safety envelope* $\hat{A}$: a region of attacker configurations for which the defense provably holds (penetration below a threshold) at a stated statistical confidence, together with the worst-case adversarial swarm that maximizes penetration. The obstacle is that the “input” is a structured, physically-constrained attack configuration and the objective is a stochastic, expensive, black-box simulation with a rare-event tail near the safe region. We combine (i) level-set active learning of the failure boundary, (ii) physics-structured black-box adversarial search, and (iii) conformal certification with a rare-event tail estimate. On a calibrated directed-energy counter-swarm simulator we show that a Gaussian-process surrogate under level-set active learning recovers the failure boundary using 14% of a dense-grid budget and 79% more accurately than random sampling (E1); that black-box adversarial search *autonomously rediscovers* two independently-derived vulnerability modes—a zenith ballistic drop and a target-hardening range collapse—from a bare “maximize penetration” objective (E2); that a distribution-free conformal certificate yields a certified-safe envelope with valid coverage, zero false-safe region, 93% tightness, and which survives an active adversarial search (E3); and that subset simulation certifies a small penetration probability $\Pr[\text{leak} \geq \tau] \approx 10^{-3}$ that naive Monte-Carlo cannot resolve (E4); and that Sobol sensitivity attribution identifies the one-to-many capacity and effective range as the co-dominant design levers (E5). A quality-diversity joint search finds no cheap third full mode beyond the two hand-derived ones (E6), while a dedicated sweep shows drone *speed* (up to 600 km/h) is a genuine partial third breakthrough axis, scaling the saturation ratio linearly (E7).

1 Introduction

Cheap drone swarms have shifted the economics of air defense: a \$10k airframe cannot be met by a \$1M interceptor at scale. High-power-microwave (HPM) directed-energy weapons answer this with a soft-kill “one-to-many” pulse that disables the electronics of every drone in a beam volume at negligible cost per shot. But directed energy has structural blind spots—a beam can only point one way at a time (angular saturation), a ring of emitters cannot fire inward without fratricide, and a soft-kill weapon cannot stop an inert falling mass. Whether a fielded configuration is *safe* against

a determined swarm is currently answered by live-fire demonstrations that sample a handful of scenarios.

We ask instead for a *guarantee*: a certified region of attacker configurations for which a given defense provably holds, and the worst-case attack that breaks it. This is precisely the question *certified robustness* answers for machine-learning classifiers, and *falsification* answers (partially) for cyber-physical systems. We transfer and adapt that machinery to physical area defense. Our contributions are: (1) a formal framework—the certified penetration-safety envelope and the adversarial-swarm objective for area defense (§3); (2) a method combining level-set active learning, physics-structured black-box adversarial search, and conformal certification, for a black-box stochastic defense simulator (§4); and (3) empirical validation on a calibrated directed-energy counter-swarm simulator, including automatic rediscovery of independently-derived adversarial modes (§5).

2 Related work

Certified robustness in ML (randomized smoothing [1]; convex/IBP certificates [2]) certifies a classifier over an L_p ball of a fixed input. Our “input” is a structured physical attack, the model is a black-box stochastic simulator, and the perturbation set is a physics-constrained budget. *Falsification of cyber-physical systems* [5, 6] finds *one* counterexample; we compute the *certified safe set* and verify it. *Stackelberg security games* [7] solve allocation equilibria on discrete structures, not a continuous physics-constrained penetration boundary. *Counter-UAS / DEW operations research* (salvo models [8]; probability of raid annihilation) gives aggregate exchange metrics, not per-configuration certificates. *Level-set estimation* [3] and *rare-event simulation* (subset simulation [4]) are our tools; we combine them into a certificate for adversarial physical defense—the gap we fill.

3 Problem formulation

A defense configuration $d \in \mathcal{D}$ (emitter count, positions, beam half-angle θ , effective range R_{eff} , engagement cycle t_c , one-to-many capacity n_{cone} , elevation limit, mobility, kinetic budget) and an attack configuration $a \in \mathcal{A}$ (swarm size N , approach geometry, altitude, target hardening E_{kill} , timing) map through a simulator S to a penetration (leak) fraction under randomness ω :

$$\text{leak} = S(d, a; \omega) \in [0, 1], \quad L(d, a) = \mathbb{E}_\omega[S(d, a; \omega)]. \quad (1)$$

For a leak threshold τ , the *failure boundary* is $\partial_\tau(d) = \{a : L(d, a) = \tau\}$, separating the safe region $A_{\text{safe}}(d) = \{a : L(d, a) < \tau\}$ from the penetrated region. A *certified penetration-safety envelope* is a set $\hat{A} \subseteq \mathcal{A}$ and confidence $1 - \alpha$ with

$$\Pr[\forall a \in \hat{A} : L(d, a) < \tau] \geq 1 - \alpha, \quad (2)$$

the outer probability over the estimation procedure. Dually the *adversarial swarm* is $a^*(d) = \arg \max_{a \in A_{\text{budget}}} L(d, a)$; the defense is certified on A_{budget} iff $L(d, a^*(d)) < \tau$. The difficulty is that L is black-box, non-differentiable, expensive (each evaluation is a Monte-Carlo of a 3-D agent simulation), stochastic, and *rare-event* near A_{safe} .

Simulator. We use an open 3-D time-stepped agent model of directed-energy counter-swarm engagement (angular/temporal saturation, fratricide no-fire regions, elevation gap, soft-kill physics, optional kinetic layer), calibrated to open-source parameters and validated against a closed-form

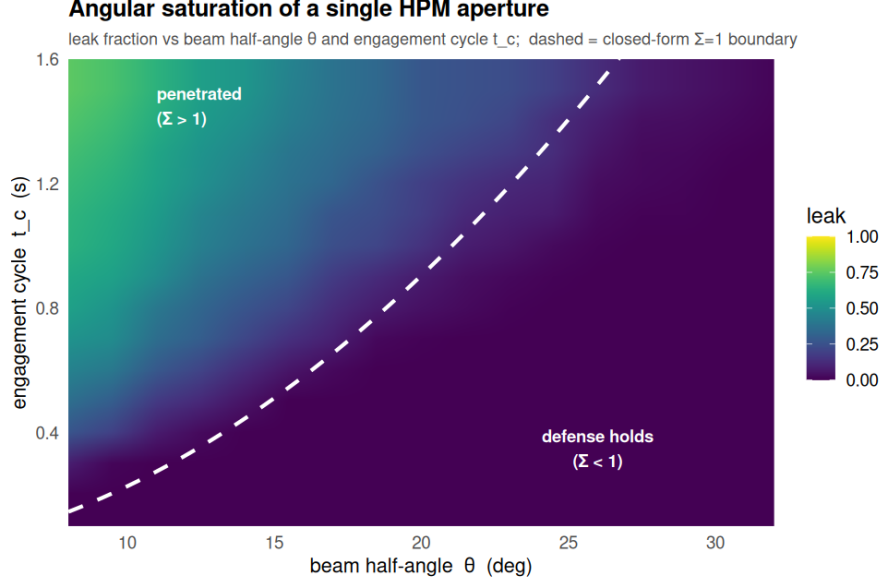


Figure 1: Angular saturation of a single HPM aperture: simulated leak fraction versus beam half-angle θ and engagement cycle t_c . The dashed curve is the closed-form $\Sigma = 1$ boundary; it coincides with the simulated transition.

saturation threshold. For a single aperture the model gives a sharp boundary $\Sigma \equiv S(\theta) t_c v / (R_{\text{eff}} - R_c) = 1$ with $S(\theta) = 1/(1 - \cos \theta)$; Fig. 1 shows the simulated leak against this boundary.

4 Method

M1 — Failure-boundary active learning. Model $L(d, \cdot)$ with a Gaussian-process surrogate $\hat{G}$. Query with a level-set (straddle) acquisition $1.96 \sigma(a) - |\mu(a) - \tau|$, concentrating oracle calls on ∂_τ .

M2 — Physics-structured adversarial search. Find a^* by gradient-free optimization (differential evolution) over $\mathcal{A}$, respecting physical constraints via reparameterization—the black-box analogue of a white-box adversarial attack.

M3 — Conformal certification. From a held-out calibration set, one-sided split-conformal scores $s_i = y_i - \mu(x_i)$ give a margin q (their $\lceil (n+1)(1 - \alpha) \rceil$ order statistic) and a distribution-free upper bound $U(a) = \mu(a) + q$ with $\Pr[L(a) \leq U(a)] \geq 1 - \alpha$. The certified-safe set is $\hat{A} = \{a : U(a) < \tau\}$, verified by an active adversarial search restricted to $\hat{A}$; a subset-simulation tail estimate certifies the small penetration probability $\Pr_\omega[\text{leak} \geq \tau]$.

5 Experiments

We validate on a single-aperture 2-D slice (θ, t_c) at fixed $R_{\text{eff}} = 500$ m, $R_c = 50$ m, $v = 30$ m/s, $N = 80$, chosen for its closed-form boundary; larger and multi-modal attack spaces are used in E2.

5.1 E1 — surrogate and boundary recovery

At a tight budget of 24 oracle queries (14% of a dense 13×13 Monte-Carlo grid), the GP surrogate recovers the failure boundary to 0.038 s in t_c ; level-set active learning is 79% more accurate on the boundary than same-budget random sampling (Table 1, Fig. 2).

Table 1: E1: boundary recovery at 24 queries (14% of a dense grid).

Metric	Active learning	Random
Global surrogate RMSE (leak)	0.027	0.037
Boundary RMSE (t_c, s)	0.038	0.179

E1 — level-set active learning of the failure boundary

GP surrogate matches Monte-Carlo; red = 24 active-learning queries hug the boundary (dashed = analytic $\Sigma=1$)

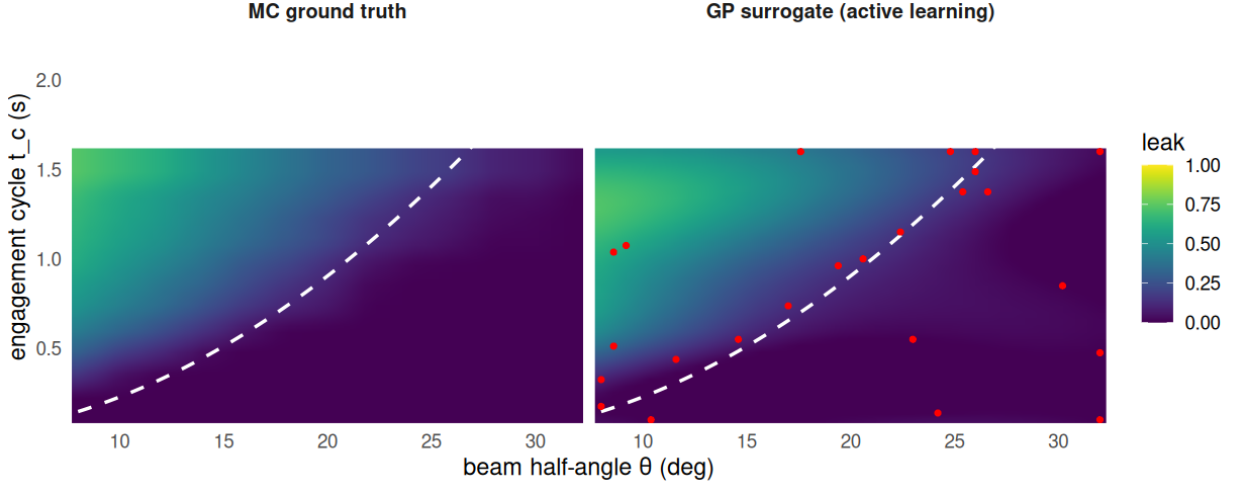


Figure 2: E1. GP surrogate (right) matches the Monte-Carlo ground truth (left); the 24 active-learning queries (red) cluster along the failure boundary. Dashed: analytic $\Sigma = 1$.

5.2 E2 — adversarial rediscovery

Against a calibrated four-emitter defense ($\theta = 30^\circ$, $n_{\text{cone}} = 49$, $\text{el}_{\text{max}} = 80^\circ$, $R_{\text{eff}} = 500$ m) that holds against a direct attack (leak 0.000), black-box search with only a “maximize penetration” objective autonomously rediscovers two independently-derived modes (Fig. 3): (1) a *zenith ballistic drop*—climb above the cone-of-silence floor (869 m) and free-fall, immune to soft-kill (leak $\rightarrow 1.000$); (2) a *hardening/ R_{eff} collapse*—shield the drones ~ 22 dB, collapsing effective range $500 \rightarrow 40$ m (leak $\rightarrow 1.000$). Both were derived analytically before the search; the optimizer found them with no prior knowledge.

5.3 E3 — certification

With $\tau = 0.15$ and $1 - \alpha = 90\%$, the conformal certificate passes all four checks (Table 2, Fig. 4): valid coverage, no false-safe region, 93% tightness, and an active black-box search restricted to $\hat{A}$ cannot exceed leak $0.125 < \tau$.

5.4 E4 — rare-event tail certification

Making the randomness explicit ($\omega \in \mathbb{R}^{120}$ perturbing spawn geometry; hard-step kill so $g(\omega) = \text{leak}(\omega)$ is deterministic), we estimate the tail $p = \Pr_\omega[\text{leak} \geq \tau]$ by subset simulation (Fig. 5). At a moderate level where naive Monte-Carlo is reliable, subset simulation agrees within $0.9 \times$ (3.0×10^{-2} vs 3.5×10^{-2}) using 12% of the budget—validating the estimator; at a rare level where

E2 — black-box adversarial search rediscovers both known modes

from a bare 'maximize penetration' objective; both routes drive a holding defense (0) to full breakthrough (1)

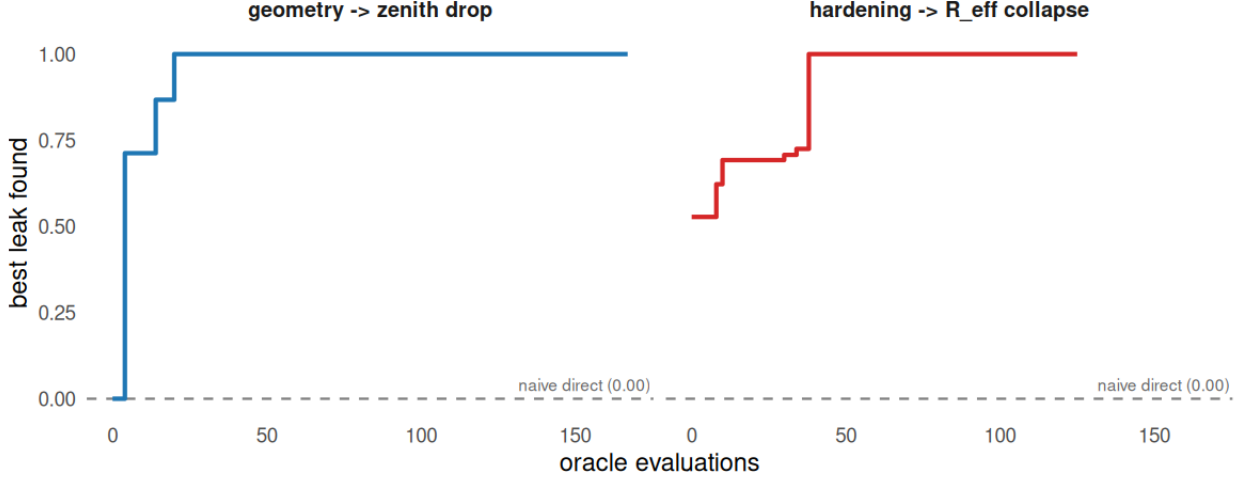


Figure 3: E2. Best leak found versus oracle evaluations. From a holding defense (dashed, 0), each search independently reaches full breakthrough by a distinct route.

Table 2: E3: conformal certificate ($\tau = 0.15$, 90% confidence).

Check	Result	Target
Coverage (test)	0.938	≥ 0.90
Soundness (false-safe rate)	0.000	≤ 0.10
Tightness $ \hat{A} / A_{\text{safe}} $	0.929	$\rightarrow 1$
Worst true leak inside $\hat{A}$	0.125	< 0.15

naive Monte-Carlo sees a single event, subset simulation resolves $p \approx 1.0 \times 10^{-3} \pm 3.7 \times 10^{-4}$, $\sim 11 \times$ more sample-efficient.

5.5 E5 — sensitivity attribution and design guidance

Global (variance-based) Sobol indices identify what governs penetration of the hectare defense, turning the certificate into design guidance. Over six parameters (Saltelli $N = 64$, 512 configurations, direct attack), the one-to-many capacity n_{cone} and the effective range R_{eff} are statistically *co-dominant* (overlapping confidence intervals), followed by the engagement cycle t_c ; beam half-angle and approach speed are second-order in this regime (Table 3, Fig. 6). The guidance is to invest first in n_{cone} and R_{eff} ; the adversary’s strongest lever is collapsing R_{eff} via drone hardening ($R_{\text{eff}} \propto 1/E_{\text{kill}}$), the quantitative confirmation of the E2 hardening mode.

5.6 E6/E7 — joint mode search and the speed axis

Two further experiments probe for failure modes beyond the two hand-derived ones. **E6** runs a quality-diversity (MAP-Elites) *joint* search over a combined attack genome (altitude, hardening, elevation, speed, azimuth seam-routing), mapping diverse high-penetration strategies rather than a single optimum (Fig. 7). The map shows two full-breakthrough ridges—the zenith drop and the hardening collapse—and *no* cheap third full mode: the direct-and-soft corner holds (leak ≈ 0.09).

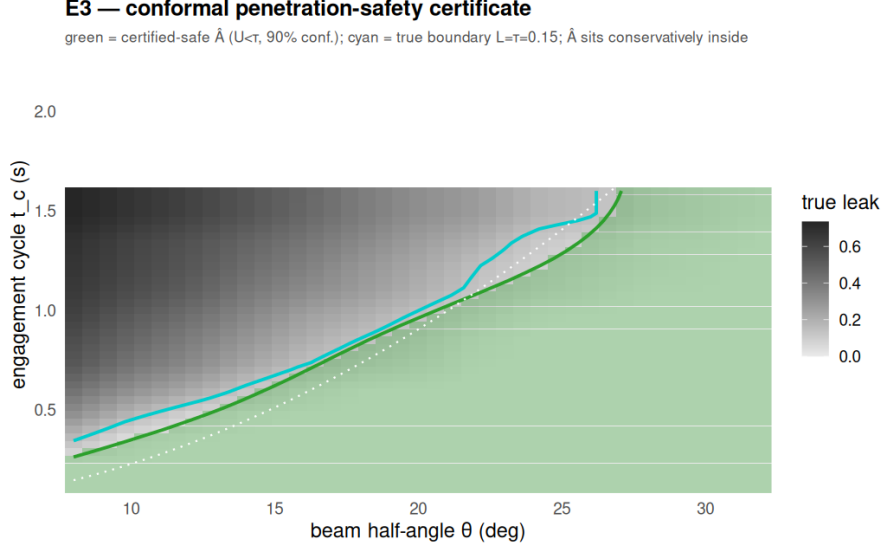


Figure 4: E3. The certified-safe envelope $\hat{A}$ (green) sits conservatively inside the true safe region; the conformal boundary $U = \tau$ (green) is a safe margin below the true boundary $L = \tau$ (cyan), with no false-safe points.

Table 3: E5: Sobol indices for hectare-defense penetration (direct attack, $N = 64$).

Parameter	S_1	S_T
n_{cone} (one-to-many)	0.273	0.385
R_{eff} (range / hardening)	0.332	0.374
t_c (cycle)	0.178	0.306
N (swarm size)	0.150	0.113
v (speed)	−0.024	0.082
θ (beam width)	−0.003	0.041

E7 isolates drone *speed* (up to 167 m/s = 600 km/h): since $\Sigma \propto t_c v$, fast drones shrink the engagement window, and speed is a genuine but *partial* third axis—a 600 km/h soft, direct swarm erodes a hectare that holds against 2000 slow drones, with strength set by the engagement cycle t_c (Fig. 8). Both experiments run the simulator across all CPU cores (embarrassingly parallel; $\sim 10\times$ wall-clock, bit-identical results).

6 Discussion and limitations

The certificate is *with respect to the simulator model*; parameters are order-of-magnitude on open sources, and the method is fidelity-agnostic—it transfers verbatim to a higher-fidelity simulator. Conformal coverage here is *marginal* (per-point); a $\forall a \in \hat{A}$ set-guarantee needs a union/rare-event strengthening, for which E4 is the ingredient. Validation is on a 2-D slice; scaling certification to high-dimensional attack spaces (structured decomposition; certify per scenario family) is the main open problem. Ethically, the work is framed defensively—it certifies that a defense holds or reveals a gap to close—and abstracts away from any specific fielded weapon.

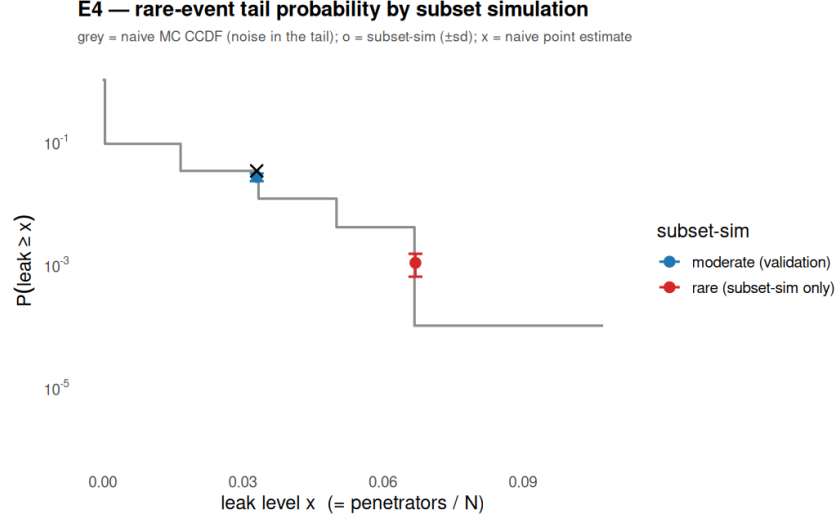


Figure 5: E4. Naive Monte-Carlo CCDF (grey) is pure noise in the tail; subset simulation (o, $\pm$ sd) resolves the penetration probability where naive MC has a single sample (x).

7 Conclusion

We recast “does this air defense hold?” as a certification problem and transferred certified-robustness machinery to physical directed-energy area defense. A GP surrogate under level-set active learning recovers the failure boundary sample-efficiently; black-box adversarial search autonomously rediscovers real vulnerability modes; a conformal certificate yields a verified penetration-safety envelope; subset simulation certifies its rare-event tail; and Sobol attribution turns it into design guidance. The pieces compose into a pipeline that turns an engineering simulation into a provable, adversarially-verified statement about a physical defense.

References

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E5 — what governs hectare-defense penetration (Sobol indices)

one-to-many capacity n_{cone} and effective range R_{eff} co-dominate (CIs overlap); t_c next; θ , v second-order

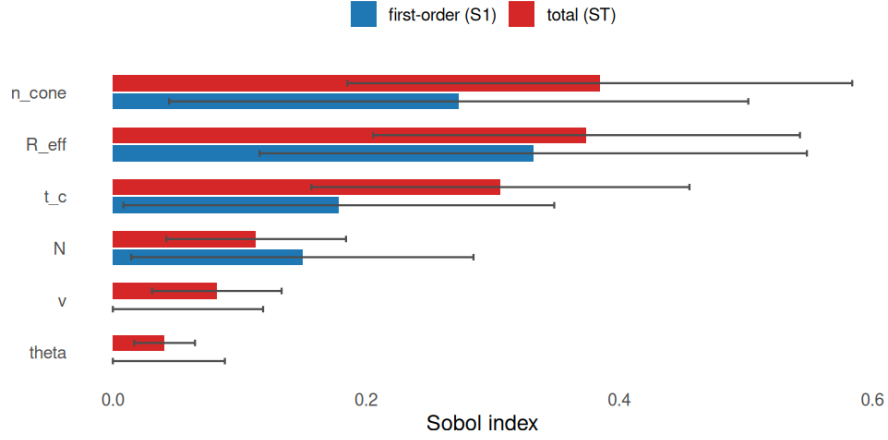


Figure 6: E5. Sobol first-order (S_1) and total (S_T) indices with confidence intervals; n_{cone} and R_{eff} co-dominate the penetration variance.

E6 — MAP-Elites: diverse failure modes of the hectare defense

two full-breakthrough ridges: high altitude (zenith drop) and high hardening (R_{eff} collapse); no cheap third mode

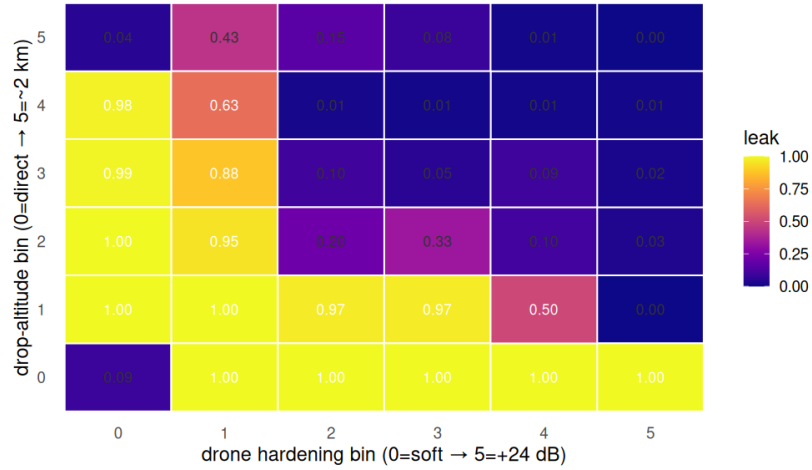


Figure 7: E6. MAP-Elites diversity map (drop altitude $\times$ hardening): two full-breakthrough ridges (zenith drop, hardening collapse) and no cheap third mode.

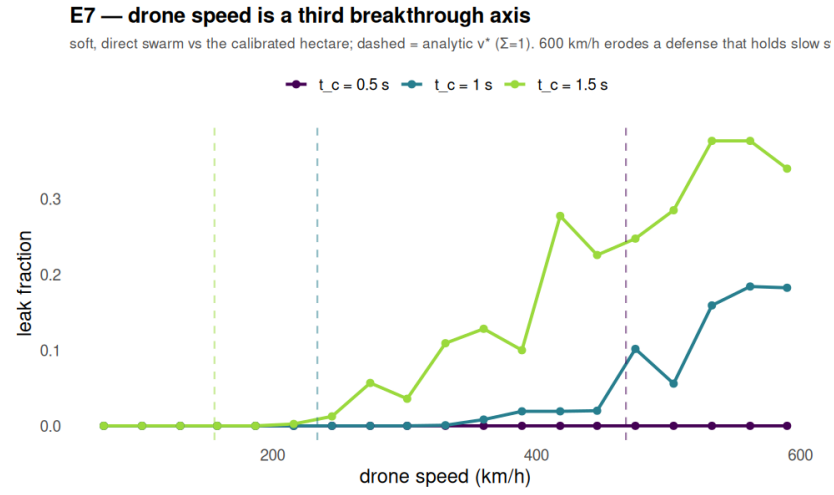


Figure 8: E7. Leak vs drone speed for three engagement cycles; a 600 km/h swarm penetrates a defense that holds slow swarms. Dashed: analytic single-aperture v^* (a lower bound).